

REVIEW

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Advances in Robotic Peg-in-Hole Assembly: A Comprehensive Review

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Abstract

As the demands for assembly quality and efficiency increase, robot-assisted assembly applications are becoming more widespread. Peg-in-hole assembly, as a typical form of assembly, has been widely researched by scholars. Currently, robotic peg-in-hole assembly faces challenges such as complex analysis of part contact forces, difficulties in task modeling, and the failure of traditional strategies. Simply controlling the position of the robot's end effector cannot achieve high precision, high efficiency peg-in-hole assembly. Flexible assembly, especially intelligent flexible assembly, is becoming the future development trend. So there is a lack of comprehensive reviews on robotic flexible peg-in-hole assembly. This paper first outlines the basic components of peg-in-hole assembly and summarizes the two basic operational processes of peg-in-hole assembly, along with their related theoretical foundations. We then review and analyze the research on passive compliant assembly, active compliant assembly, and intelligent flexible assembly. Finally, it presents an outlook on the future development directions of robotic peg-in-hole assembly.

Keywords Robot, Peg-in-hole assembly, Small clearance, Motion planning, Compliant assembly

1 Introduction

Assembly is the process of integrating components into complete products and verifying their compliance through testing, constituting approximately 40% of total production costs. With increasing demands for assembly efficiency and quality, particularly driven by advancements in artificial intelligence (AI) and smart manufacturing technologies, robotic assembly has emerged as a critical trend in future industrial development [1–3]. Peg-in-hole assembly, a prevalent assembly paradigm, faces limitations when traditional position-based control strategies are applied to small-clearance, high-precision

components. This process necessitates adaptive adjustments of relative poses between components based on real-time contact forces, yet encounters challenges including dynamic force coupling, difficulties in modeling complex geometric constraints, and insufficient adaptability to unstructured environments. In certain scenarios, human intervention remains indispensable, significantly hindering the widespread adoption of robotic assembly technologies [4, 5].

Although notable progress has been achieved in passive compliance devices (PCDs), active impedance control (AIC), and intelligent learning algorithms, persistent bottlenecks exist in cross-modal perception fusion [6] and robustness in dynamic environments [7]. For instance, existing compliant assembly strategies are predominantly scenario-specific, lacking generalizability [8], while intelligent algorithms heavily rely on extensive simulation data with inefficient virtual-to-real migration [9]. Consequently, it is of significant theoretical and practical value for advancing robotic intelligent manufacturing by systematically organizing the technological framework of

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robotic peg-in-hole assembly, analyzing core challenges, and proposing interdisciplinary innovation pathways.

Current reviews on robotic peg-in-hole assembly predominantly focus on the perception-decision-regulation-collaboration technical architecture of robotic assembly systems, emphasizing coupled relationships among multidimensional technologies through discussions on robotic assembly methodologies [10–12], application domains [13–15], and human-robot collaboration [16, 17]. However, a comprehensive review systematically grounded in mechanical analysis of peg-in-hole operations remains absent. This paper conducts a review of robotic peg-in-hole assembly, including encompassing system composition, operational theory, compliance strategies, and future trends. Section 2 dissects core components and their interaction mechanisms in peg-in-hole assembly systems. Section 3 investigates dynamic modeling and contact-state optimization methods during search-alignment and insertion phases. Section 4 comparatively analyzes the technical characteristics and limitations of passive compliance, active compliance, and intelligent flexible assembly. Section 5 outlines future directions in ultra-precision assembly, green transformation, and interdisciplinary integration, with their logical relationships visualized in Figure 1. The parts selected by the red dotted line are the key points of this thesis. By consolidating multidisciplinary research achievements, this work aims to provide systematic technical references

for academia and industry while establishing theoretical foundations for intelligent breakthroughs in complex assembly scenarios.

2 Peg-in-Hole Assembly Systems

Based on the research of robotic peg-in-hole assembly tasks, this study divides the assembly process into three sequential phases: the search phase, alignment phase, and insertion phase [18]. However, current technological investigations predominantly focus on the joint optimization of the search and alignment phases. Consequently, these two phases are consolidated into a unified search-alignment phase for technical analysis in this work. Given the diverse geometric characteristics and mating conditions of peg-hole components across different assembly tasks, each presents unique technical challenges. A peg-in-hole assembly system can generally be divided into three critical subsystems: assembly workpieces, manipulators, and sensing components [19]. Through coordinated interactions among these subsystems, robotic systems enhance performance and efficiency in complex assembly tasks to achieve successful peg-in-hole assembly, as illustrated in the accompanying Figure 2 [20].

2.1 Assembly Workpieces

Modern manufacturing exhibits trends toward high precision, diversified specifications, and rapid product iteration [21], leading to increasingly heterogeneous

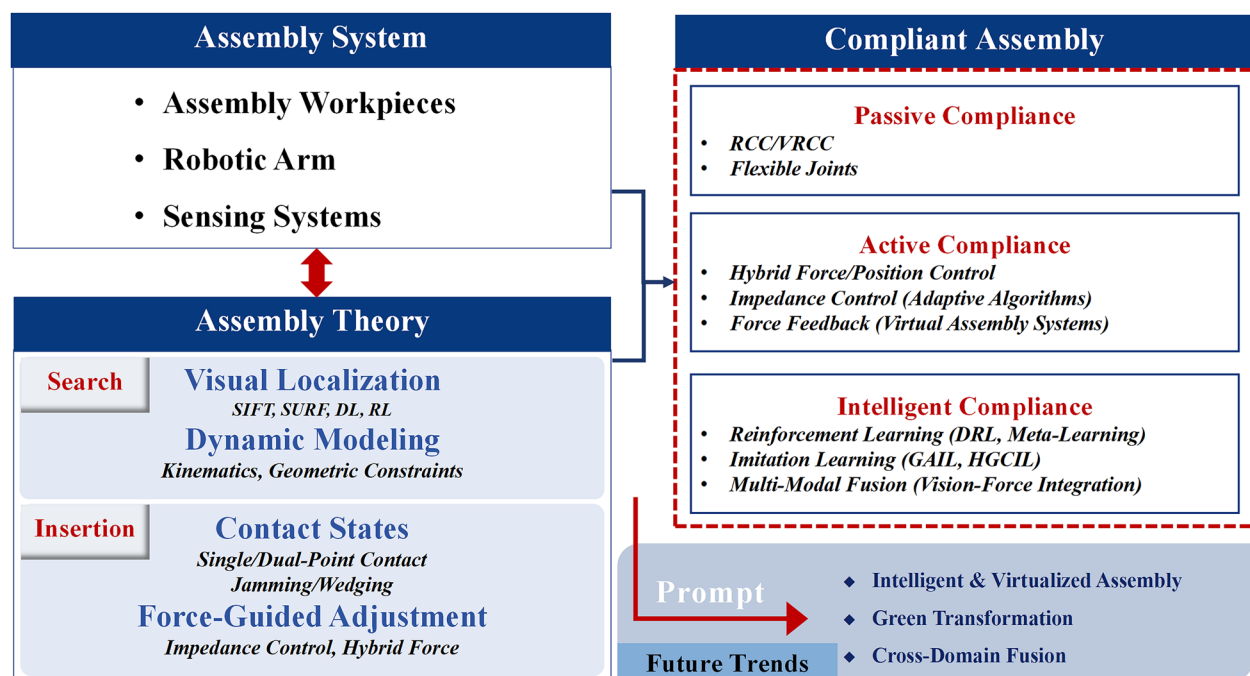


Figure 1 Framework of the review

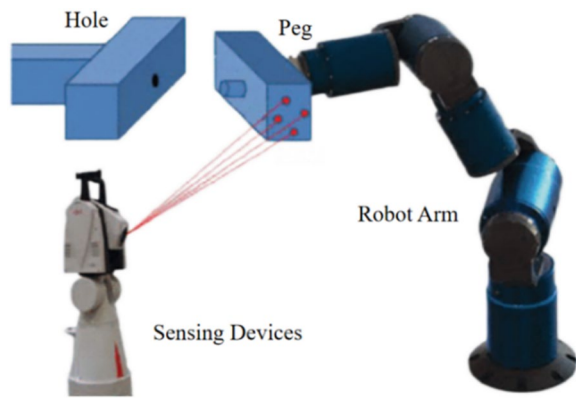


Figure 2 Peg-in-hole assembly system

morphologies of assembly workpieces. Assembly workpieces are typically categorized into peg components (grasped and precisely adjusted by robotic manipulators) and hole components (fixed on workstations as assembly targets). They commonly include the types of cylindrical peg-in-hole assemblies [22], square peg-in-hole assemblies [23], and irregular peg-in-hole assemblies [24]. Based on the number of peg-hole pairs, these tasks can be further classified into single-peg-hole (SPH) [25], dual-peg-hole (DPH) [26], and multi-peg-hole (MPH) [27] assemblies. Notably, the geometric dimensions and fit tolerances of peg-hole pairs directly determine task difficulty. Generally, assembly challenges escalate with increased geometric complexity, higher numbers of mating pairs, and reduced assembly tolerances. In industrial applications, cylindrical assemblies with keyed pins have gained widespread adoption [28]. However, such small-clearance, high-precision tasks face issues including complex contact constraints, difficulties in task modeling, and potential failure of traditional strategies, prompting extensive research. For example, Gan et al. [29] proposed an online constraint domain generation method for complex part assembly, utilizing primitive convex shape combinations to approximate non-convex geometries. Wu et al. [30] developed a novel assembly strategy for non-convex components combining circular and rectangular features.

2.2 Robotic Manipulators

Robotic manipulators serve as the core actuation systems for peg-in-hole assembly tasks. Precise motion control enables manipulators to grasp, position, and ultimately insert pegs into holes. Industrial or collaborative robotic arms typically feature 6 degrees of freedom (DOF) or 7 DOF configurations [31]. Compared to 6-DOF manipulators, 7-DOF systems offer enhanced operational flexibility and safety. For grasping operations, uncertainties

arising from dynamic open-service environments impose stringent demands on gripping performance. Mainstream end-effectors include: tendon-driven two-finger grippers (TDFGs) [32], direct-drive three-finger grippers (DDTFGs) [33], and soft five-finger dexterous hands (SFDHs) [34].

In industrial settings, TDFGs and DDTFGs dominate due to their suitability for rigid, regular-shaped components, while SFDHs exhibit superior versatility but face limited adoption owing to structural complexity and high costs [31]. Addressing these limitations, Ueda et al. [35] developed a compact gripper integrated with a differential mechanism, enabling efficient small-clearance peg-in-hole assembly without complex control or hole-searching procedures.

2.3 Sensing Components

Sensing systems constitute critical components for acquiring state information during assembly processes and guiding manipulator motions, primarily categorized into vision sensors and force sensors [36].

- (1) Vision Sensors: These include 2D cameras and 3D cameras. 2D cameras, mounted on workstations, enable hole localization by identifying hole features in images, subsequently driving manipulator movements near the target hole via position servo or visual servo. However, 2D cameras fail to capture complete depth information at close ranges due to environmental noise interference [37].
- (2) Force Sensors: Force sensors are subdivided into joint torque sensors (integrated into robotic arms) and six-axis force/torque (F/T) sensors (mounted on manipulator end-effectors as needed) [38].

3 Peg-in-Hole Assembly Operations

Based on the aforementioned peg-in-hole assembly system, this study further investigates robotic operational workflows in peg-in-hole tasks, which are divided into two phases: search-alignment phase and insertion phase [18]. In human-operated assembly, operators typically perceive geometric features (e.g., key points or critical edges) on workpiece surfaces, establish contact and alignment through these features, and finally apply appropriate assembly forces to complete the task. Similarly, robotic workflows prioritize feature localization through visual or tactile representations, significantly simplifying assembly procedures. Despite substantial variations in peg-hole geometries, most workpieces exhibit distinct mating edges that enable rapid localization via assembly-specific semantic features, thereby determining their positions and poses.

Feature representation methodologies continue to evolve in peg-in-hole assembly. For instance, Sun et al. [39] proposed a neural network-based heatmap representation for keypoints, demonstrating superior spatial generalization compared to direct pixel coordinate regression. This work focuses on general cases of peg-hole feature analysis, particularly emphasizing mating edge selection strategies by examining contact edges from both ends to dissect assembly dynamics.

Current peg-in-hole assembly research predominantly relies on virtual or physical testbeds. Nemec et al. [40] developed the "Reconcell" autonomous robotic workstation, enabling skill transfer (e.g., polishing and grinding) through human demonstration and automated task planning for assembly cells. Niekum et al. [41] designed a robotic system for assembling IKEA furniture, enhancing production flexibility and balancing productivity with adaptability. While these advancements expand the functional capabilities of assembly systems, their operational workflows diverge from the peg-in-hole framework discussed in this study. Accordingly, this work initiates its analysis from the search-alignment phase to establish a universal theoretical foundation for robotic peg-in-hole operations.

3.1 Search-Alignment Phase

The search-alignment process involves robotic grasping and positional alignment of components. In small-clearance assembly tasks, robots rarely achieve insertion in a single attempt, necessitating iterative search operations. To prevent component damage from excessive forces during insertion, closed-loop control via sensor feedback is essential for continuous measurement and correction. As introduced earlier, widely studied sensors include vision sensors and force sensors, with vision sensors typically employed first to estimate hole positions and poses.

In industrial production, vision serves as a crucial role in enabling robots to perceive their environment. Vision-guided localization offers non-contact operation, high efficiency, and rapid dynamic response, significantly enhancing robotic flexibility while providing real-time, non-contact measurement data to support assembly strategies. During manipulator operations, visual localization of hole components is required to guide robotic motion toward target holes, where localization accuracy directly impacts subsequent assembly efficiency and success rates [42]. Extensive research has focused on workpiece visual localization in assembly, with methodologies broadly categorized into three classes: artificial feature-based visual localization, fiducial marker-based visual localization, and deep learning-based visual localization [31].

3.1.1 Artificial Feature-Based Visual Localization Methods

Artificial feature-based methods achieve high-precision localization by designing specialized image processing algorithms to extract geometric features (e.g., edges, corners, contours) and performing coordinate transformations. The core workflow includes image preprocessing, feature extraction, feature matching, and pose estimation. For instance, line segmentation and CAD model matching are common approaches for objects with sharp edges [24]. In scenarios with blurred edges or low contrast, feature extraction algorithms with translation, rotation, and scale-invariant properties become critical. Algorithms such as scale-invariant feature transform (SIFT) [43], speeded up robust features (SURF) [44], and oriented FAST and rotated BRIEF (ORB) [45] enable robust feature detection across varying scales and illumination conditions, finding widespread applications in edge localization.

Recent advancements address industrial-specific requirements through algorithm refinements. For example, Mikolajczyk et al. [46] replaced SIFT keypoints with Harris corners and constructed 32-dimensional feature descriptors using Gaussian circular windows, significantly improving industrial image matching efficiency—though this method primarily excels in indoor/outdoor environments. Li et al. [47] enhanced real-time localization performance in SURF by modifying feature point dimensionality, achieving high accuracy in assembly processes.

However, artificial feature-based methods require meticulous parameter tuning for different assembly scenarios. When handling complex workpieces, limited feature robustness and susceptibility to mismatches frequently lead to significant localization errors [48].

3.1.2 Fiducial Marker-Based Visual Localization Methods

In practical assembly tasks, due to factors such as environmental lighting, workpiece shape, and material texture, vision-based localization methods relying on handcrafted features may struggle to accurately obtain the position and pose of the workpiece. However, marker-based vision localization methods can effectively address this issue. In robot peg-in-hole assembly, marker-based vision localization methods achieve high-precision pose estimation through predefined physical or virtual markers, with the core concept being the construction of a stable spatial mapping relationship using the geometric properties or encoded information of the markers. Plane target markers such as ARTag [49], AprilTag [50, 51], and ArUco [52, 53] are widely used in various assembly tasks due to their simple deployment, stable feature recognition, and high pose estimation accuracy. In recent years,

researchers have further improved localization robustness through multi-sensor fusion, optimized marker design, and detection algorithms. For example, Zhang et al. [54] proposed a multi-camera collaborative calibration method, utilizing a distributed stereo vision system combined with a marker-based stereo matching algorithm. By optimizing the baseline distance and field-of-view matching strategy between cameras, they reduced the 3D reconstruction error to within 0.1 mm. Shao et al. [55] designed encoded markers based on structured light projection, using plane targets with two (or more) concentric circle patterns to complete line-structured light vision sensor (LSLVS) calibration. In a 200×200 mm field of view, the accuracy of the calibration method achieved was 0.07 mm.

To address the limitations of monocular vision, Chen et al. [56] proposed a dynamic localization framework that integrates an inertial measurement unit (IMU) with markers. This framework uses the IMU to compensate for image distortion caused by motion blur in the robotic arm and quickly matches marker contours through an improved version of the generalized Hough transform (GHT)—the hierarchical probabilistic Hough transform (HPHT). Experimental results showed that the localization speed was improved by 40% compared to traditional methods. In large-scale assembly scenarios, Wang et al. [57] combined laser tracking and multi-view markers, proposing a hybrid localization model. By jointly optimizing pose calculations with laser ranging data and the spatial geometric constraints of visual markers, they achieved absolute positioning accuracy within ± 0.05 mm for large workpiece assembly.

Additionally, adaptive marker selection and dynamic updates have become a research hotspot in recent years. For instance, Liu et al. [58] proposed a dynamic marker selection algorithm based on random sample consensus (RANSAC). By real-time evaluation of marker visibility and stability, they dynamically removed occluded or severely deformed markers, maintaining over 90% localization success even in partially occluded scenarios. Huang et al. [59] designed deformable markers for high-dynamic assembly environments, combining elastic materials with fluorescent coatings to ensure stable detection through optical flow tracking, even when the marker moves rapidly with the robotic arm.

3.1.3 Deep Learning-Based Visual Localization Methods

Deep learning-based vision localization methods achieve high-precision positioning in robot peg-in-hole assembly through data-driven feature extraction and end-to-end pose estimation. The core idea is to use deep neural networks (DNNs) to automatically learn robust feature representations in complex environments and directly map

them to assembly actions or pose parameters. Compared to vision localization methods based on handcrafted features, deep learning techniques, particularly convolutional neural networks (CNNs), can automatically learn representative features through dataset training, eliminating the need for manual feature design. In recent years, researchers have significantly enhanced the generalization and stability of these methods by improving network architectures, integrating multimodal data, and optimizing training strategies. For instance, in systems with unknown impedance modeling parameters and uncertain contact environment models, Tang et al. [60] constructed an error model and proposed an adaptive fuzzy neural network control scheme based on impedance learning. This method enables the robot to perform impedance adaptive control for adjustment goals generated by impedance learning without prior knowledge of uncertainties or a sufficient amount of observational data, achieving autonomous correction of peg-in-hole assembly deviation with a gap of 0.04 mm and a hole diameter of 25.4 mm. Chen et al. [61] employed a multi-scale attention fusion network (MAFNet), which concurrently extracts local texture features and global semantic information from images, integrating 3D point cloud data to generate high-precision peg-in-hole pose estimates. Liu et al. [62] introduced a robot grasping target detection algorithm based on synthetic data, using domain randomization techniques to quickly and accurately generate training samples.

When features of targets are collected through video or in real-time via cameras, they are typically categorized into static descriptors extracted from the main frame, descriptors from video objects, and motion descriptors. As mentioned earlier, SIFT is one of the most important descriptors in content-based video classification [63], and using SIFT descriptors for image and video classification yields high accuracy. However, SIFT descriptors are applied to independent frames and ignore the motion of the object itself. Lowe et al. [43] introduced a local invariant feature tracking (LIFT) descriptor that tracks SIFT descriptors across continuous frames in each shot, accounting for the dynamics of the video, thus enhancing the robustness and stability of feature points. To reduce reliance on annotated data and use unlabeled image data for feature point detection and description, Detone et al. [64] proposed superpoint, a self-supervised learning-based feature point detection and description method, utilizing two decoder networks to represent key points and descriptors, respectively. To further improve the accuracy of feature matching, Sarlin et al. [65] introduced an end-to-end feature matching method, superglue, based on attention mechanisms and graph neural networks. Based on this, Gao et al. [66] proposed a

technical solution using superpoint for feature extraction and superglue for front-end feature matching. By integrating this into the VINS-Fusion system, they improved visual localization accuracy in complex environments, significantly reducing the system's loop closure error along the Y-axis from 0.40 m to 0.17 m. Due to the high image matching precision of superpoint and superglue, and the robustness of local feature matching with transformers (LoFTR) in areas with poor texture, Hoshi et al. [67] combined superglue, superpoint, and LoFTR to improve the accuracy of camera parameter estimation in structure from motion (SfM) and reconstruction precision in multi-view stereo (MVS).

There are also cases where positioning accuracy is improved by adjusting the number, performance, posture, and image dimensions of the cameras: Chang et al. [68] utilized two monocular cameras, extracting the four edge points of a USB square hole based on corner detection and image thresholding, and then determined the center position of the hole; Xu et al. [69] used a combination of a remote global camera and a proximal hand-eye camera, performing initial localization using an improved NCC template matching algorithm. When the robotic arm moves close to the hole, the proximal camera performs a secondary localization based on edge features, effectively reducing positioning errors; Zhang et al. [70] proposed the random-verify generalized Hough transform (RV-GHT), which constructs a more concise shape dictionary on a single image compared to the traditional generalized Hough transform [71], significantly improving algorithm performance. Based on deep learning, workpiece feature points can be output and integrated with the robotic arm's motion, directly producing assembly actions. Triyonoputro et al. [72] used images stitched from two hand-eye cameras, recognizing the relative position between the peg-in-hole using neural networks, and discretizing it into directions represented by four quadrants, guiding the robotic arm to perform stepwise motion in that direction; Haugaard et al. [73] used multiple cameras to simultaneously estimate the center positions of the peg and hole, driving the robotic arm for continuous visual servo to improve assembly efficiency. However, the aforementioned methods are only suitable for circular peg-in-hole tasks and do not apply to workpieces of other geometric shapes. To address this issue, Puang et al. [74] proposed the KOVIS visual servo method, which uses one neural network to learn key-point representations from images, and another network to learn the robot's motion based on keypoints from the images, enabling use for workpieces of various shapes.

In fact, for most peg-in-hole assembly tasks, where the exact position of the hole is not determined by manual instructions, a global camera is usually installed in the

environment or a hand-eye camera is mounted at the end of the robotic arm to assist with localization. The global camera is used for large-scale transfer of the peg, while the hand-eye camera guides the peg closer to the hole. When the peg is very close to the hole, the camera image is often blocked and cannot provide effective information. In such cases, feedback from force sensors is needed to further adjust the position and posture of the robotic arm to complete the assembly task [75]. The summary table of visual positioning techniques in the search and alignment stage is shown in Table 1. Among them, the numbers 1 to 6, 7 to 9, and 10 to 15 are respectively visual positioning methods based on benchmark markers, artificial features, and deep learning.

3.2 Insertion Phase

During the insertion phase of peg-in-hole assembly, the hole's preliminary position and pose are estimated using the localization methods outlined in Section 2.1. The manipulator then approaches and contacts the hole via motion planning to initiate insertion. Due to inevitable errors in vision sensors and robotic systems, perfect alignment between the peg and hole is rarely achieved at this stage. Furthermore, visual sensors lose efficacy due to occlusion, necessitating F/T sensor feedback to refine the peg's position and orientation. Tailored assembly algorithms and strategies are applied based on workpiece characteristics to ensure smooth insertion.

The insertion process involves analyzing F/T data to infer the peg-hole relative position and orientation, thereby determining contact states and adjustment directions. Contact forces and moments are strongly coupled with peg position and orientation, reflecting the inherently three-dimensional dynamic nature of peg-in-hole assembly [76]. To simplify spatial force and position constraints, contact states are categorized into three types: Chamfer contact state (initial alignment via chamfer guidance), Single-point contact state (partial misalignment), Two-point contact state (dual-contact stabilization).

Starting from a visually coarse-aligned chamfer contact state, the system transitions toward two-point contact through force-guided adjustments. Force sensors integrated into the robotic end-effector provide real-time feedback during grasping and insertion. For clarity, this section uses cylindrical peg-in-hole assembly as a representative example.

The workpiece undergoes several fundamental state transitions from the gripping operation to the final assembly, including non-contact states, collision and impact processes, and continuous contact motion. These states can be represented by a unilateral constraint model, including the non-contact state, impact state,

Table 1 Summary table of visual positioning technologies

	Technology method	Key characteristics	Representative research	Major innovation
1	Multi-camera Collaborative Calibration	Stereo Vision & Marker Matching	Zhang et al. [54]	Optimized camera baseline and field-of-view, reducing 3D reconstruction error to within 0.1 mm
2	Encoded Markers with Structured Light	LSLVS	Shao et al. [55]	Achieved calibration accuracy of 0.07 mm in a 200×200 mm field of view
3	Dynamic Localization Framework	IMU & Marker Fusion	Chen et al. [56]	Uses IMU to compensate for motion blur, improving localization speed by 40%
4	Hybrid Localization Model	Laser Tracking & Vision Fusion	Wang et al. [57]	Achieved absolute positioning accuracy within ±0.05 mm for large workpiece assembly
5	Dynamic Marker Selection Algorithm	Dynamic Selection & Occlusion Resistance	Liu et al. [58]	Dynamically removes occluded or deformed markers, maintaining over 90% success in partially occluded scenarios
6	Deformable Markers	Deformable Markers for High Dynamics	Huang et al. [59]	Ensures stable marker detection in high-dynamic environments where the robotic arm moves rapidly
7	SIFT	Scale-Invariant Feature Transform	Mikolajczyk et al. [46]	Using Harris corners and Gaussian circular window feature descriptors
8	SURF	Speeded Up Robust Features	Li et al. [47]	Modified feature point dimensionality, improved real-time localization performance
9	ORB	Oriented FAST and Rotated BRIEF	-	Applicable to different scales and illumination conditions
10	Adaptive Fuzzy Neural Network	Unknown Impedance Modeling	Tang et al. [60]	Peg-in-hole assembly deviation correction, gap of 0.04mm
11	Multi-scale Attention Fusion Network	Multi-modal Data	Chen et al. [61]	Combining 3D point cloud, high-precision pose estimation
12	Synthetic Data Domain Randomization	Target Detection	Liu et al. [62]	Rapid generation of training samples
13	New Feature Matching Method	Attention Mechanism	Sarlin et al. [65]	SuperGlue end-to-end feature matching
14	Multi-camera Collaboration	Pose Estimation	Haugaard et al. [73]	Simultaneous estimation of peg and hole center positions
15	Shape-Independent Method	Universal Localization	Puang et al. [74]	KOVIS visual servo, applicable to various geometric shapes

Table 2 Fundamental state transitions and their descriptions of workpiece

Fundamental state transitions	State descriptions
No-contact state	$g_n > 0$
Contact impact state	$g_n = 0, \dot{g}_n > 0$
Continuous stable contact state	$g_n = 0, \dot{g}_n = 0, \dot{g}_t \neq 0$
Static contact state	$g_n = 0, \dot{g}_n = 0, \dot{g}_t = 0$

continuous stable contact state, and stationary contact state. These states are described as shown in Table 2. This section does not consider the impact issues that arise from initial contact. The symbol g represents the distance relationship between two contact bodies, g_n represents

the displacement of the peg during the assembly process, and $\dot{g}_n$ and $\dot{g}_t$ represent the normal and tangential constraint velocities, respectively.

The contact states defined in this paper are all point-contact forms, with the contact motion mode being sliding contact without rolling. The unilateral constraint condition defined by the constraint distance function can be expressed in the form of column vectors, with its Jacobian matrix g_p was given by Eq. (1). This matrix is used to represent the projection mapping of the contact force and torque in the coordinate space, where p is the pose vector.

$$g_p = \partial g / \partial p. \quad (1)$$

For dynamic assembly systems, the constraint conditions change over time and are expressed in a complete constraint form. When contact occurs, the constraint dynamics equation between the assembly body and the wall can be expressed as Eq. (2),

$$M\ddot{\mathbf{p}} = \mathbf{F} + (\mathbf{g}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t) + \mathbf{f}_{\mu})\boldsymbol{\lambda} = \mathbf{F} + \mathbf{G}\boldsymbol{\lambda}, \quad (2)$$

where $\mathbf{M}$ represents the inertia matrix; $\mathbf{F}$ represents the input external force; $\boldsymbol{\lambda}$ denotes the Lagrange multipliers, which indicate the contact constraint forces in the constrained system, essentially representing the composition of microscopic deformation forces; $\mathbf{f}_{\mu}$ is the vector of tangential deformation constraint forces; $\mathbf{g}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t)$ represents the Jacobian matrix of the constraint force due to deformation in the coordinate space; $\mathbf{G}$ is expressed by Eq. (3),

$$\begin{aligned} \mathbf{G} &= \mathbf{g}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t) + \mathbf{f}_{\mu} \\ &= \mathbf{g}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t) + \mathbf{h}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t)\boldsymbol{\lambda} \\ &= \mathbf{g}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t) - \mu(\text{sgn}v_t)\boldsymbol{\lambda}, \end{aligned} \quad (3)$$

where $\mathbf{h}_{\varepsilon\mathbf{p}}^T(\mathbf{p}, t)\boldsymbol{\lambda}$ is the matrix that is "linearly related to the friction coefficient"; v_t represents the tangential velocity at the contact point. Eq. (2) is converted into matrix form,

$$\begin{pmatrix} \mathbf{M} & -\mathbf{G} \\ \mathbf{g}_{\varepsilon\mathbf{p}} & 0 \end{pmatrix} \begin{pmatrix} \ddot{\mathbf{p}} \\ \boldsymbol{\lambda} \end{pmatrix} = \begin{pmatrix} \mathbf{F} \\ \mathbf{a} \end{pmatrix}, \quad (4)$$

where $\mathbf{a}$ denotes the normal contact acceleration. Assuming the matrix on the left-hand side is invertible, the accelerations $\ddot{\mathbf{p}}$ and constraint contact deformation forces $\boldsymbol{\lambda}$ can be solved from the above equation. Based on the degree of constraint and the redundant equations of the independent coordinates, the above equation can be decomposed and simplified. The pose vector $\mathbf{p}$ can be decomposed into the independent coordinates $\mathbf{p}_i$ and the constrained coordinate quantities $\mathbf{p}_j$; similarly, $\mathbf{g}_{\varepsilon\mathbf{p}}$ can be decomposed into the corresponding $\mathbf{g}_{\varepsilon\mathbf{p}_i}$ and $\mathbf{g}_{\varepsilon\mathbf{p}_j}$, and by ensuring that $\mathbf{g}_{\varepsilon\mathbf{p}_i}$ is full rank, the constraint dynamics equation can be represented as Eq. (5).

$$(\mathbf{M}\mathbf{A} - \mathbf{G}) \begin{pmatrix} \ddot{\mathbf{p}}_i \\ \boldsymbol{\lambda} \end{pmatrix} = \mathbf{F} - \mathbf{M} \begin{pmatrix} 0 \\ \mathbf{g}_{\varepsilon\mathbf{p}_j}^{-1} \end{pmatrix}. \quad (5)$$

After obtaining the constraint motion equation, we use the two-dimensional peg-in-hole model analysis method proposed by Whitney [77], selecting the intersection plane of the peg-in-hole axis for force analysis. We then establish the assembly contact model and further propose the following assumptions:

- (1) Rigidity assumption: During the assembly process, the contact forces between the peg-in-hole are small, and the physical shape of the peg-in-hole does not change. Only the fixture part experiences minor compliant deformation. Both the peg and hole components are considered rigid, with independent lateral and torsional stiffness.
- (2) Quasi-static assumption: The robot's motion speed is slow, and the inertia effects due to acceleration are neglected. The assembly process is considered to be in a quasi-static equilibrium state.
- (3) Friction assumption: The dynamic friction coefficient is equal to the static friction coefficient. During the chamfering stage, the force analysis is similar to a point contact scenario, mainly focusing on the relationship between force and displacement for single-point and two-point contacts.

After visual coarse positioning, the peg-in-hole enters a point contact state, as shown in Figure 3 [18].

At this point, the peg experiences normal force N_1 and frictional force f_1 , where $f_1 = \mu N_1$ and μ represent the friction coefficients. The hole radius is R ; the peg radius is r ; l_p is the distance from the peg end to the sensor; l is the distance from the force application point to the sensor; M is the torque at the sensor location, and θ is the angular deviation of the peg-in-hole, also

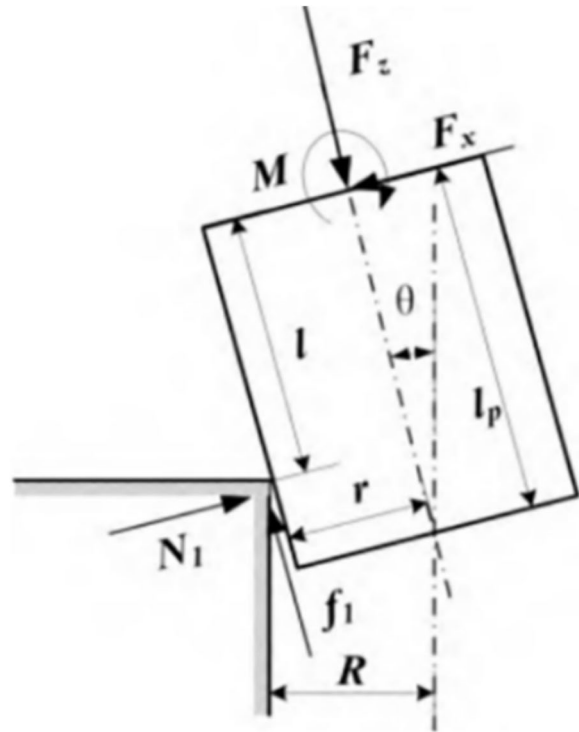


Figure 3 Single-point contact state

referred to as the pose angle. A force analysis is conducted to obtain the force and torque on the force sensor, which are represented by Eq. (6).

$$\begin{cases} F_x = N_1, \\ F_z = f_1 = \mu N_1, \\ M = -f_1 r - N_1(l_p - l). \end{cases} \quad (6)$$

To avoid jamming phenomena and ensure smooth assembly of the peg into the hole, an assembly force $F = [F_x, F_z, M]^T$ needs to be applied to the assembly body. The positive directions of the components of the assembly force are consistent with the diagram. When the peg contacts the left side, the angle of deviation of the peg center relative to the hole axis is less than 0 when it is offset to the left. In this state, the next adjustment step should be performed, and this fine-tuning distance must satisfy the physical property equation shown below, as represented by Eq. (7).

$$\begin{bmatrix} F_x \\ F_z \\ M_\theta \end{bmatrix} = \begin{bmatrix} K(U - U_0) \\ \mu F_x \\ K_\theta(\theta - \theta_0) \end{bmatrix}, \quad (7)$$

where U is the offset of the peg relative to the hole axis; θ_0 is the angular deviation relative to the hole; K is the lateral stiffness; K_θ is and the torsional stiffness. After entering the point contact position, the peg continues to assemble downward. As the peg further inserts, the bottom of the peg will get closer to and eventually touch the

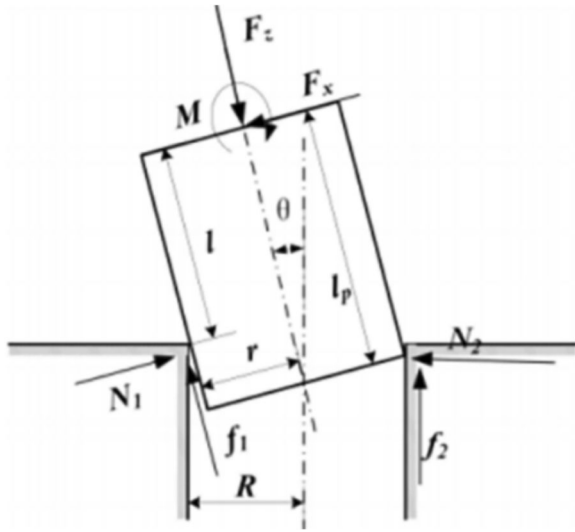


Figure 4 Two-point contact state

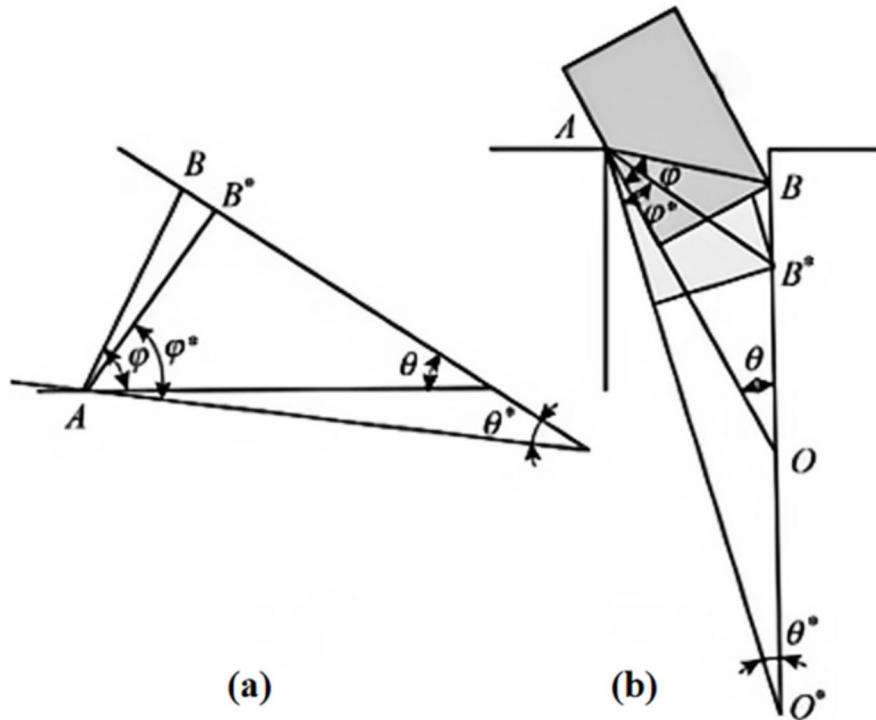


Figure 5 Contact states: **a** Two-point contact state simplified and **b** Concrete model

inner wall of the hole, reaching a two-point contact state, as shown in Figure 4 [18].

In the two-point contact state of the peg-in-hole, as illustrated in Figure 5 [18]. The right side of the peg experiences normal force N_2 , and frictional force f_2 , which satisfies the force balance equation as represented by Eq. (8).

$$\begin{cases} F_x = N_1 - N_2 \cos \theta + f_2 \sin \theta, \\ F_z = f_1 + N_2 \sin \theta + f_2 \cos \theta, \\ M = -f_1 r - N_1(l_p - l) + (N_2 \sin \theta + f_2 \cos \theta)r. \end{cases} \quad (8)$$

The corresponding fine-tuning equation is given by Eq. (9).

$$\begin{bmatrix} F_x \\ F_z \\ M_\theta \end{bmatrix} = \begin{bmatrix} K(U - U_0) \\ \mu N_1 + N_2 \sin \theta + \mu N_2 \cos \theta \\ K_\theta(\theta - \theta_0) \end{bmatrix}, \quad (9)$$

During the adjustment process, the angular adjustment is determined by the torque. This strategy categorizes the torque into two types: the first type is the interference torque generated by the contact force caused by position deviation, and the second type is the desired torque caused by angular deviation. These two torques will be coupled and superimposed, affecting the judgment. Considering the peg-in-hole assembly task, when the peg is inserted into the hole at a certain tilt angle θ , multiple states exist whether it is in a point contact or two-point contact. Figure 5 shows the two-point contact state and its simplified model at different insertion depths of the peg into the hole. Assuming the contact end of the peg is cylindrical and the wall surface is flat, the contact deformation area will be elliptical. Point $A(x_a, y_a)$ and point $B(x_b, y_b)$ are the left and right contact points of the peg-in-hole.

In the case of peg-in-hole assembly with small clearance, the insertion process inevitably faces jamming and wedging problems. During the dynamic insertion process, due to the imbalance between the input F/T ratios, a jamming state may occur. This is mainly due to two situations. First, the direction of the inertial force falls within the friction cone, causing the system to be unable to move. Second, the direction of the inertial force coincides linearly with the boundary of the friction cone and the reaction force, which causes both the inertial force and the reaction force to tend toward infinity [18]. This jamming state is similar to the phenomenon of impact. Wedging, on the other hand, is a static geometric phenomenon that occurs in systems with two or more constraints. It is caused by the system's binding force, which is linearly related to geometric deformation under the action of an external

force [77]. According to the Coulomb friction model, the frictional force resists the occurrence of slippage, and when the acceleration generated by the frictional force tends toward infinity, its direction will inevitably be opposite to the direction of the system's motion. At this point, the sliding speed rapidly oscillates to zero, leading to a jamming state. Therefore, we take matrices S and T as the orthogonal matrices of MA and G in Eq. (5), as shown in Eqs. (10) and (11).

$$S(MA) = 0 \quad (10)$$

$$TG = 0 \quad (11)$$

As long as matrix $T(MA)$ is a singular matrix, the system will experience a jamming phenomenon. When matrix SG is a singular matrix, the force tends toward infinity, and the wedging state occurs. We will now analyze each of these scenarios separately.

3.2.1 Single-Point Jamming Analysis (Using Point A in Figure 5(b) as an Example)

To analyze and verify the jamming state based on dynamic analysis, we consider single-point contact at point A: considering the conditions with deformation present, according to the dynamic Eq. (12), we obtain

$$\begin{pmatrix} m & 0 & 0 & \mu \operatorname{sgn} v_x \\ 0 & m & 0 & -1 \\ 0 & 0 & J & L_A \\ 0 & 1 & -L_A \cos \varphi & 0 \end{pmatrix} \begin{pmatrix} \ddot{x} \\ \ddot{y} \\ \varphi \\ \lambda \end{pmatrix} = \begin{pmatrix} F_x \\ F_y \\ M \\ \lambda \end{pmatrix}, \quad (12)$$

where m and J are the coefficients related to the inertia matrix; $L_A = (l - \delta_A)(\cos \varphi + \mu \operatorname{sgn} v_x \sin \varphi)$; and l represents the half distance between points A and B; δ_A is the deformation at point A. When matrix $T(MA)$ is a singular matrix, the minimum value of the sliding friction coefficient when the jamming occurs at point A can be solved, as shown in Eq. (13).

$$\mu_1 = -\operatorname{sgn} v_x \left(\frac{J + m l_A^2 \cos^2 \varphi}{m l_A^2 \sin \varphi \cos \varphi} \right). \quad (13)$$

When the sliding friction coefficient reaches the minimum value and $\varphi \in (0, \pi/2)$, $v_x < 0$, the jamming phenomenon will occur.

3.2.2 Two-Point Jamming Analysis (Using Points A and B in Figure 5(b) as an Example)

To analyze and verify the wedging state based on dynamic analysis, we consider two-point contact at points A and B. Considering the conditions with deformation present, we obtain Eq. (14),

$$\begin{pmatrix} M \\ \mathbf{g}_p \end{pmatrix} = \begin{pmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & m \\ 0 & 1 & -l_A \cos \varphi \\ \sin \theta & \cos \theta & (l - \delta_B) \cos(\varphi + \theta) \end{pmatrix}, \quad (14)$$

where δ_B represents the deformation at point B . When matrix $\mathbf{SG}$ is a singular matrix, and assuming $\delta_A \approx \delta_B$, the sliding friction coefficient at the A and B points can be solved when the wedging state occurs. The conditions for wedging occurrence are shown in Eqs. (15) and (16). Based on experimental analysis, the space for successful assembly without jamming and the decreasing probability of wedging as the equipment's load capacity and material elasticity increase are derived [18].

$$v_{Ax} < 0, v_{Bx} > 0, \quad (15)$$

$$\mu_2 = (1/\tan \varphi \ 1/\tan \varphi). \quad (16)$$

Whether addressing the jamming or wedging problem, both are analyzed based on the Coulomb friction model and the single-contact constraint model. The conditions for jamming and wedging occurrence are derived from the singularity of the coefficient matrix in the equation system. In fact, the two-dimensional peg-in-hole model proposed by Whitney [77] can more clearly resolve the

jamming and wedging problems, providing a schematic diagram of the conditions to avoid assembly jamming, as shown in Figure 6 [76].

This diagram mainly normalizes Eq. (5), setting $\lambda = (l_p - l)/(2r\mu)$ the parallelogram to represent the feasible assembly region. If the points formed by the main F/T are located within this region, jamming will not occur; conversely, jamming will occur. In the robotic peg-in-hole assembly process, both jamming and wedging have been important topics of academic research and industrial applications. Many scholars have proposed various solutions and methods to address the common jamming and wedging problems in peg-in-hole assembly. Through experimental research and theoretical analysis, these methods have significantly improved the precision and stability of peg-in-hole assembly. For example, regarding the jamming problem, Xie et al. [78] developed a jamming detection technology based on visual perception that uses friction thermodynamics to analyze the temperature rise pattern and combines ANSYS friction simulation to determine the criteria for jamming. Li et al. [79] developed a jamming detection technology based on force feedback sensors by collecting vibration signals from key areas and extracting typical signals, proposing a target state monitoring and fault diagnosis method. Regarding the wedging problem, Yao et al. [80] used

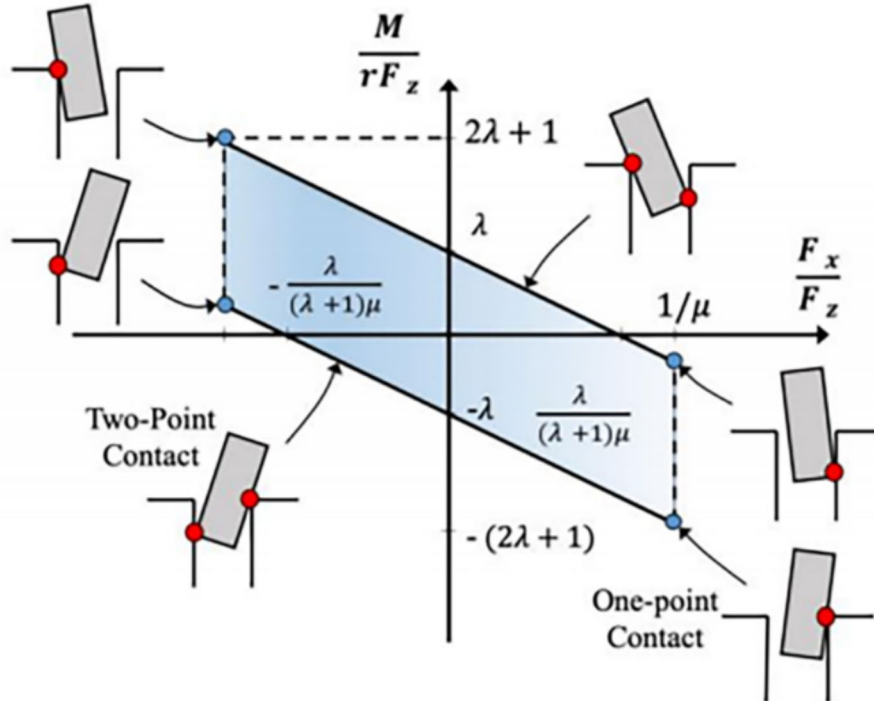


Figure 6 Two-dimensional peg-in-hole model

axial thrust tooling for thrust testing of guide bars, utilizing DEFORM-3D finite element software to numerically simulate the wedging process of the guide bar and evaluate its wedging quality, effectively predicting the wedging risk during the assembly process. These methods have not only significantly improved the precision and efficiency of peg-in-hole assembly but have also provided new ideas for the development of compliant assembly technology. However, as the complexity of peg-in-hole workpieces and the diversification of the assembly process continue to increase, how to further optimize intelligent solutions to address jamming and wedging problems remains a key research direction in the future.

4 Compliant Assembly

In Section 3, we conducted an in-depth analysis of the dynamic characteristics of the peg-in-hole assembly process, with a particular focus on the fundamental principles of jamming and wedging phenomena. However, with the goal of improving production efficiency and reducing labor costs, it is especially important to advance robots' ability to perform high-precision, complex assembly tasks. This is not only a key measure for enterprises to enhance their level of automation, but also requires the establishment of a reliable feedback mechanism during long-duration, high-load assembly processes, while ensuring system stability and high reliability [81, 82]. Early studies mainly focused on the design and optimization based on the characteristics of the robot assembly model itself, often using techniques such as chamfering [83] and variable stiffness actuators [84–86] to optimize the assembly process. With the rapid development of sensor technologies, sensor-based assembly methods have gradually gained widespread application in the field of robotic assembly. This technological advancement holds significant theoretical value and practical importance in enhancing automation levels and advancing intelligent manufacturing.

In the mechanical arm peg-in-hole assembly process, due to the inevitable contact forces between the mechanical arm and the environment, excessive contact forces could potentially cause damage to the workpiece or equipment. Therefore, compliant assembly technologies have been introduced to regulate the peg-in-hole contact forces. To address the issue of compliant assembly for robotic arms, both domestic and international scholars have conducted a substantial amount of related research, which can be broadly categorized into three main methods: passive compliant assembly, active compliant assembly, and intelligent flexible assembly.

The following section will provide a detailed explanation of these three methods.

4.1 Passive Compliant Assembly

In certain high-precision assembly tasks, flexible wrist devices are widely used to reduce the uncertainty in robotic assembly systems. The remote compliance center (RCC), first developed by Whitney [77], is a typical flexible wrist device and is generally installed between the robot's end-effector and the assembly component. This device consists of a six-degree-of-freedom spring mechanism, which adjusts the stiffness of the springs to meet different compliance requirements. During the assembly process, the deformation of the springs can effectively compensate for peg-in-hole pose errors, with the spring's elastic constant directly determining the stiffness of the compliant device. Although the RCC offers an intelligent solution to address the peg-in-hole problem, its center is fixed, requiring different RCCs for targets of varying lengths [87]. In response to this limitation, scholars have made various improvements: Lee [88] introduced the concept of a variable RCC (VRCC), which incorporates an elastomer shear pad (ESP) as a new design element for the RCC, simplifying its mechanical structure to make it suitable for use in actual assembly lines. Young et al. [89] developed a miniature assembly system composed of a micro gripper, RCC unit, voice coil motor drive mechanism, and precision motion platform, designed for assembling small parts. To reduce assembly costs, many researchers have also designed RCC devices with different DOF: Jiang et al. [90] proposed a 3-degree-of-freedom RCC planar compliant mechanism to compensate for shifts caused by positioning errors and dimensional tolerances; Jain et al. [91] introduced a new active 4-degree-of-freedom RCC mechanism that compensates for the positional and directional errors between two mating parts during the robot peg-in-hole assembly process.

With the rapid development of electromechanical technology and component miniaturization, flexible wrist device designs are becoming increasingly refined to adapt to more complex and diverse assembly tasks and reduce the complex calibration processes required by RCC-like compliant devices. Liu et al. [92] proposed a biomimetic flexible mechanical wrist (BFMW) to passively compensate for assembly errors or component damage caused by charging assembly errors during the connection process. Tanaka et al. [93] proposed a cable-driven flexible wrist and developed a new transfer reinforcement learning (RL) method, transfer learning by aggregating dynamics models (TRANS-AM), enabling soft robots to complete target task assemblies with fewer steps. Flexible joints based on robotic arms are also used for passive compliant

assembly: Campbell et al. [94] utilized the Baxter robot's series elastic actuator (SEA) joints [95] to achieve passive compliance in the assembly process, enabling collaborative work with large objects.

4.2 Active Compliant Assembly

Active compliant assembly refers to the real-time feedback of contact force information between the robotic arm and the environment to the control system, enabling the robotic arm to actively adjust its motion based on the contact situation, thereby achieving compliance with both the environment and the object being manipulated. Compared to passive compliant assembly, active compliant assembly offers greater versatility and applicability, allowing it to be flexibly used in different types of robotic tasks. Active compliant assembly can be mainly classified into three methods: force-position hybrid control, impedance control, and force feedback control.

4.2.1 Force/Position Hybrid Control Methods

The force-position hybrid control method, proposed by Craig et al. [96], aims to decompose the robotic arm's motion space into two independent subspaces—force and position—and design controllers in each subspace, thereby achieving precise tracking control of both force and position. This method performs excellently in peg-in-hole assembly, particularly in dynamic assembly processes, where it can effectively adjust contact forces and peg-in-hole pose errors. Cao et al. [97] established a new terminal sliding mode and designed an adaptive position/force control law to overcome the uncertainty of the dynamic model. Chen et al. [98] proposed three peg-in-hole assembly strategies based on a selective compliance articulated robot arm (SCARA). These strategies simulate manual assembly with a hybrid switching force/position control scheme, with a visual-force sensing non-linearity of approximately 3% across the entire range. The proposed robot assembly strategies are suitable for peg-in-hole assembly in industrial scenarios. Considering the rapidly growing demand for contact-intensive tasks, Shi et al. [99] proposed a push-based hybrid position/force assembly skill that maximizes the utilization of environmental constraints during task execution, achieving a 100% success rate. Ning et al. [100] introduced a fuzzy variable admittance control (FVAC) model for precise posture estimation and adjustment of the peg.

4.2.2 Impedance Control Methods

Impedance control was first proposed by Hogan et al. [101], unifying position control and force control into a single framework and introducing techniques for controlling manipulator behavior. These techniques led to the development of a unified method for motion-constrained

movement, dynamic interaction, goal acquisition, and obstacle avoidance. Based on this, Nagchaudhuri et al. [102] initially used passive impedance control, attempting to use digital measurements of contact forces from a single robot for AIC in deburring and assembly tasks, with RCC applied to perform peg-in-hole assembly tasks. Currently, several scholars have made improvements to impedance control and applied it to various assembly scenarios: Cho et al. [103] introduced impedance control using F/T sensors to estimate the actual position of the task object and avoid excessive contact forces. They proposed a connector assembly strategy performed by industrial robots; Jing et al. [104], addressing the challenges of fine assembly of multi-point coupled electrical connectors under vision-limited conditions, proposed a self-adaptive assembly method for circular electrical connectors based on a "two-point contact-flexible sliding pose—posture correction—alignment insertion" approach, introducing a F/T-position hybrid impedance control algorithm. This provided support for spacecraft assembly and maintenance; He et al. [105] proposed a new strategy for large workpiece assembly tasks based on a hybrid impedance control algorithm, achieving compliant motion and direct force control during the insertion process to ensure even contact between the workpiece and the surface of the slot; Wang et al. [106] developed a peg-in-hole assembly control technology combining vision and force feedback to enhance compliance control of objects. Their proposed fuzzy sliding mode joint impedance controller (FSMJIC) addressed the problem of chatter in traditional sliding mode control designs while achieving the desired joint compliance. However, impedance control typically requires a reference model to achieve stable control of the robot, which necessitates precise environmental modeling or identification, limiting its adaptability in unstructured environments [75]. To address this issue, Yan et al. [9] proposed an adaptive meta-policy learning (AMPL) algorithm with a set of virtual models to obtain generalized models for multi-class assembly skills.

4.2.3 Force Feedback Control Methods

Force feedback assembly control strategies involve detecting force and torque signals during the assembly process to indirectly understand the assembly environment, and then adjusting the robotic arm's posture to compensate for peg-in-hole deviations [107]. This strategy is highly applicable in peg-in-hole assembly, especially in scenarios with minimal peg-in-hole damage and high levels of intelligence. However, traditional force feedback assembly strategies typically discretize the assembly process into multiple contact states, performing state recognition and control based on complex models.

These strategies often have high algorithmic complexity and lower recognition success rates in complex environments, requiring strong modeling capabilities to be applied in high-precision peg-in-hole assembly scenarios [108, 109]. As a result, most force feedback assembly control strategies are implemented through simulations in virtual environments.

In recent years, researchers have made several improvements to force feedback assembly strategies: Cheng et al. [110] developed a virtual assembly system with force feedback, exploring a force feedback-based virtual assembly navigation method and proposing a constraint-driven incremental algorithm to solve the problem of component separation during constrained motion; Huang et al. [111] designed and constructed a distributed collaborative virtual assembly system with 3D force feedback, where participants use force feedback devices to select appropriate assembly parts and control the parts to move in different directions, enabling multi-user tactile collaborative assembly; Sagardia et al. [112] conducted an evaluation study using virtual assembly to verify two different collision feedback modes, revealing the significant and notable superiority of force feedback in virtual assembly scenarios; Chen et al. [113] proposed a method that combines visual feedback and force feedback to reduce positioning errors in high-precision assembly tasks, showing that the assembly success rate was improved compared to the impedance control method.

4.3 Intelligent Flexible Assembly

Intelligent flexible assembly transcends traditional active compliance by integrating AI and multimodal perception, reducing dependency on structured environments and enhancing adaptability in complex scenarios. Core methodologies include RL, imitation learning (IL), multimodal data fusion, and physics-guided hybrid intelligence, all of which have achieved significant theoretical and practical progress [114–116].

4.3.1 Generalization and Decision Optimization via RL

RL is a machine learning method based on trial-and-error mechanisms, where agents explore actions in an environment and receive feedback to learn optimal strategies. Using RL for high-precision decision-making is a common approach in industrial robotic arm assembly scenarios. Liu et al. [117] proposed a model-based meta-reinforcement learning approach for robotic industrial insertion, achieving online adaptation of assembly tasks through real-time updates of dynamic models by interacting with the environment. Li et al. [118] introduced a self-imitation RL method for flexible flat cable (FFC) assembly goals, which does not rely on specific end-effectors, enabling the assembly of flexible targets. Parzeller

et al. [119] combined RL with the rapidly-exploring random tree* (RRT*) path planning algorithm for assembly sequence planning.

Compared to RL, deep reinforcement learning (DRL) integrates deep neural networks with RL, enabling better handling of complex tasks and large datasets. DRL, through autonomous exploration and interactive learning strategies, demonstrates unique advantages in unstructured environments, and its generality has led to widespread applications in device insertion assembly. Zhu et al. [120] proposed a DRL-based method for automated assembly planning in robot-based prefabricated construction. Liu et al. [121] introduced a novel DRL-based hybrid visual servoing (DRL-HVS) controller for motion planning in assembly tasks under visual servoing. However, the high interaction costs of DRL are sometimes unacceptable, and the trained models have weak generalization capabilities, making it difficult to quickly adapt to complex and changing assembly environments [26, 122]. To address this, Liang et al. [123] introduced hindsight experience replay (HER) to enable DRL to learn from failures and proposed a progress intrinsic motivation-based HER (P-HER) method to overcome these issues, applied to intelligent robots performing typical industrial tasks. Lee et al. [124] studied a F/T-based supervised learning framework "PolyFit", which represents the paradigm shift from RL to supervised learning methods. They also proposed a sim-to-real adaptation method for real-world applications to adjust the posture of the peg to correct misalignment, with an assembly success rate of over 90%, highlighting the robustness and adaptability of the proposed method.

4.3.2 Multimodal Enhancement and Generalization Extensions in IL

IL is a robot training method based on RL, where robots learn complex tasks by observing and mimicking the actions of humans or other agents. IL has significant application value in compliant assembly, especially in assembly processes that require high precision, flexibility, and adaptability. The primary goal is to enable robots to learn the operator's actions, create task or action models, and then reproduce them. IL is generally divided into two methods: task model-based learning and action model-based learning. Compared to RL, which requires extensive interaction with the environment during the training process, IL only requires a small amount of expert demonstration data to learn assembly strategies. However, its generalization capability is often limited by the size of the data [125]. Ankile et al. [126] proposed a method to improve IL performance with a smaller manual demonstration budget, applied to assembly tasks that require precise grasping, repositioning, and insertion of multiple

parts over long distances and through several task stages. Wang et al. [127] introduced a hierarchical IL framework (hierarchical goal-conditioned IL, HGCIL) based on simulated learning, solving complex robotic assembly tasks in human demonstrations by combining trajectory learning and force learning.

IL performs well in small-scale assembly processes, but limited teaching data results in a lack of generalization over a broader range. Additionally, some researchers have combined IL and RL to study different stages of the assembly process. To address the inefficiency of the traditional generative adversarial IL (GAIL) method, which suffers from low-quality generated samples leading to high interaction costs between the agent and the environment, Cai et al. [128] proposed a new method called hindsight transformation generative adversarial IL (HT-GAIL), based on deep learning. This method accelerates the convergence of the algorithm and achieves assembly success rates of 96%, 88%, and 61% for fit gaps of 1.12, 0.80, and 0.52 mm, respectively.

4.3.3 Intelligent Decision-Making via Multimodal Perception and Fusion

Multimodal data fusion enhances the decision accuracy of assembly strategies by integrating visual, tactile, and proprioceptive information. With the development of deep learning, neural networks can process high-dimensional multimodal data to generate low-dimensional feature vector representations, enabling intelligent, flexible assembly to leverage richer perceptual information. Hwang et al. [129] compared the performance of neural networks processing multimodal data (including images, F/T, and robot posture information) using multi-layer perceptron (MLP) structures and cross-attention structures. They demonstrated the effectiveness of cross-attention in improving the performance of multimodal data processing tasks in robotic assembly with different data types. Paul et al. [130] extracted intent from voice dialogues and fused the intent with any detected matching gestures used during interaction with the robot. Their deep learning-based dual intent and entity transformer (DIET) classifier was used to evaluate interactive robots in contextual assembly scenarios.

In fact, due to the more complex geometric and physical constraints typically encountered in peg-in-hole assembly tasks, existing RL methods are often difficult to apply to multiple peg-in-hole problems. To effectively address these issues, Chen et al. [131] proposed learning multiple modalities, including vision, proprioception, and F/T, into compact representations to handle complexity and uncertainty, thereby improving sample efficiency. They successfully implemented multiple peg-in-hole assemblies and generalization across different

object shapes in real-world scenarios. Duan et al. [132] studied the integration of virtual reality (VR) and remote robotic operations, aiming to facilitate collaboration between humans and robots in executing peg-in-hole assembly tasks. The introduced semi-autonomous framework utilizes visual object detection technology to identify the object's type and 6D pose, providing users with new functionalities.

In robot assembly, the peg-in-hole contact states are complex. Excessive contact forces can damage the assembly, leading to task failure, and uncertain part poses can cause low efficiency and poor environmental adaptability in robotic assembly. Therefore, designing effective assembly strategies that use sensor information and the geometric data of the assembly to ensure successful robot assembly remains a challenge in this field. Flexible contact between the workpiece and the assembly must be achieved, robot control strategies need to be improved, and more flexible designs should be adopted to enhance adaptability and operational stability in complex environments. This will drive the further development of industrial automation [76]. A more comprehensive assembly technology diagram is shown in Table 3, which summarizes the technical details of the three types of assembly: active compliance, passive compliance, and intelligent compliance, respectively. Specific technological approaches, performance metrics, challenges and limitations have all been presented.

5 Outlook

Robotic peg-in-hole assembly, as one of the core technologies of intelligent manufacturing, has made significant advancements in precision and automation. Modern robotic systems commonly integrate technologies such as laser measurement, visual recognition, and tactile feedback. Through AI and machine learning, robots are able to autonomously complete various assembly tasks, handle complex assembly scenarios, and ensure precise fitting between the peg and the hole. Future developments will deeply integrate trends such as intelligence, flexibility, high precision, and sustainability, while being profoundly influenced by national policy guidance. Key challenges in peg-in-hole assembly tasks include improving robot movement safety and efficiency, enhancing the deployment efficiency of workpiece visual positioning, and increasing the compliance of contact operations. Designing effective robot operation strategies, integrating cutting-edge technologies across fields, and expanding into emerging scenarios are crucial for enhancing the stability, robustness, and adaptability of robot operations. Based on these considerations, future research on robot peg-in-hole assembly can be explored in the following three directions [21, 133], as shown in the overall

Table 3 Summary table of flexible assembly technology

Category	Specific technological approaches	Performance metrics	Challenges and limitations
Passive compliant assembly	- Six-DOF RCC [77] - Variable RCC with Elastomer Shear Pad [88] - 3-DOF Planar Compliant Mechanism [90] - 4-DOF Active RCC Mechanism [91] - Biomimetic Flexible Mechanical Wrist [92] - Cable-Driven Flexible Wrist [93] - SEA Joints [94, 95]	- Position Error Reduction - Mechanical Compliance Range - Stiffness Adjustability - Assembly Success Rate - Miniaturization Capability	- Fixed Compliance Characteristics [87] - Limited Adaptability - Design Complexity - Environment-Specific Calibration - Mechanical Fabrication Precision
Active compliant assembly	- Force-Position Hybrid Control [96] - Adaptive Position/Force Control [97] - Hybrid Switching Force/Position Control [98] - Push-based Hybrid Position/Force Assembly [99] - Fuzzy Variable Admittance Control [100] - Impedance Control Framework [101] - F/T Sensor-Based Impedance Control [103] - Multi-point Connector Assembly [104] - Hybrid Impedance Control [105] - Vision-Force Feedback Control [106] - Virtual Assembly with Force Feedback [110] - Distributed Collaborative Virtual Assembly [111] - Visual-Force Feedback Combination [113]	- Contact Force Precision - Motion Tracking Accuracy - Adaptive Response Time - Environmental Adaptation Index - Visual-Force Sensing Nonlinearity	- Computational Complexity [75] - Precise Environment Modeling Required - Sensor Integration Challenges - Real-Time Processing Overhead - Lower Recognition Rates [108, 109]
Intelligent flexible assembly	- Model-based Meta-Reinforcement Learning [117] - Self-Imitation Reinforcement Learning [118] - DRLAssembly Planning [120] - DRL Hybrid Visual Servoing [121] - Hindsight Experience Replay [123] - Supervised Learning Framework [124] - HGIL [127] - Hindsight Transformation Generative Adversarial IL [128] - Multimodal Data Fusion with Cross-Attention [129] - Intent Extraction and Gesture Fusion [130] - Multimodal Representation Learning [131] - VR-Integrated Robotic Operations [132]	- Learning Efficiency - Generalization Capability - Sample Efficiency - Task Complexity Handling - Cross-Domain Adaptability - Assembly Success Rates	- High Interaction Costs [26, 122] - Limited Generalization - Data Dependency - Computational Resource Requirements - Difficulty in Applying to Multiple Peg-in-Hole Problems [131]

framework in Figure 7, and the dashed box selection part represents that the content within the box is a whole, and the blue text is a further refinement of the black text theme.

5.1 Development of Intelligent and Virtualized Ultra-Precision Assembly Technologies

Ultra-precision assembly represents a critical technological frontier, driven by Industry 5.0 and advanced AI methodologies [134, 135]. The core innovation lies in high-fidelity simulation techniques that generate

comprehensive digital environments, bridging the gap between simulated and real-world assembly processes. By creating massive, contextually diverse training datasets, researchers can develop more adaptive and generalized assembly strategies.

Multimodal sensor technologies revolutionize traditional assembly processes [136] by integrating visual, tactile, and haptic sensing with advanced algorithms. Graph Neural Networks (GNNs) and quantum mechanics-inspired computational approaches [137, 138] enable real-time, precise process optimization. Biomimetic

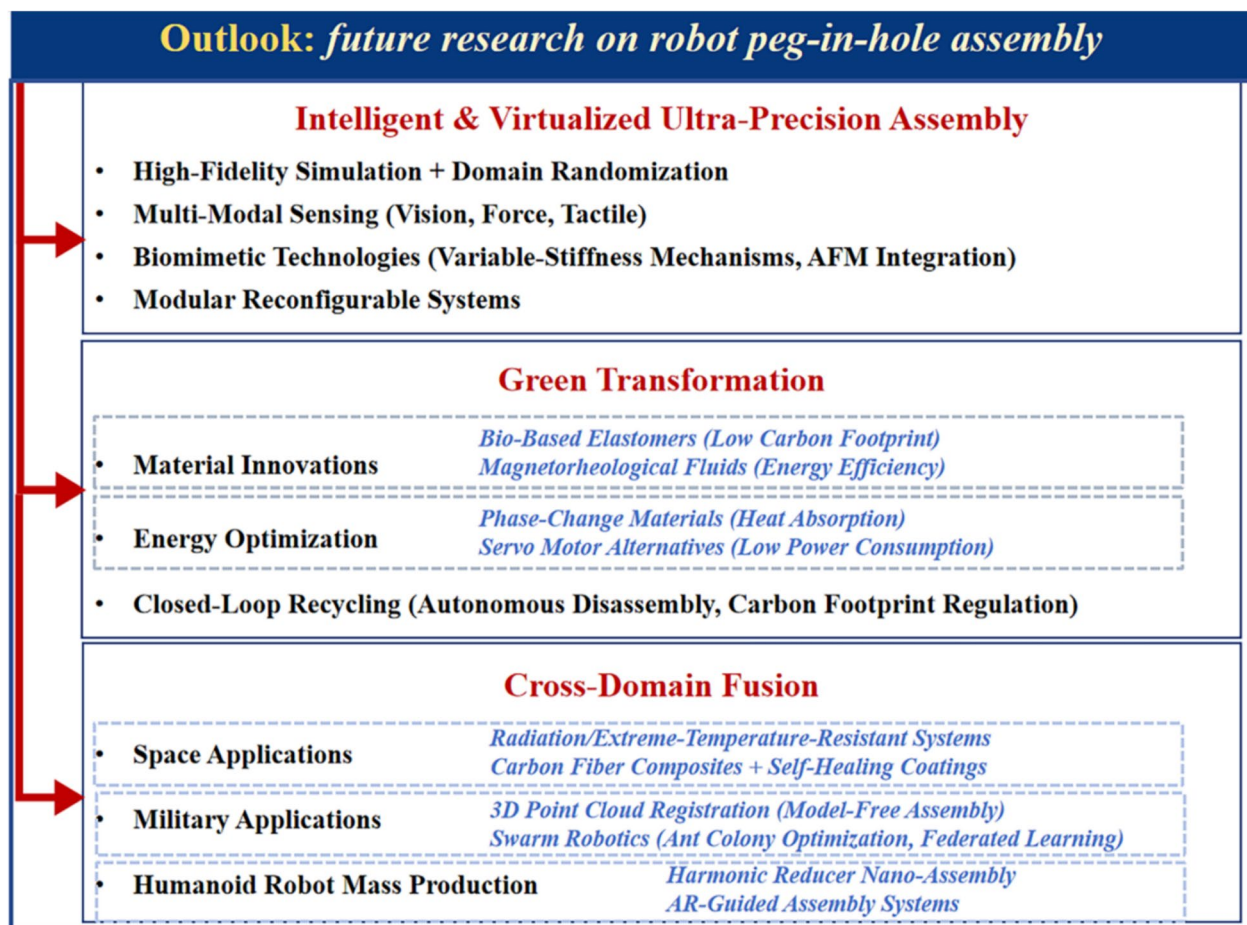


Figure 7 Outlook of the overall framework

technologies further enhance this approach by emulating biological joint characteristics [139], developing adaptive wrist devices with sophisticated stiffness-switching capabilities.

The integration of atomic force microscopy (AFM) [140] and ultra-precision grinding techniques [141] allows for microscopic deformation detection, dramatically improving assembly accuracy. Modular reconfigurable systems provide flexible solutions for multi-variety, high-precision production, expanding application prospects in high-end electronics, aerospace, and precision instruments [142].

5.2 Technology Development Under Social Environmental Issues and Policy Guidance

China's "Made in China 2035" and the "15th Five-Year Plan" emphasize green development as a critical path to sustainable manufacturing [143]. European strategies similarly focus on transitioning to a circular economy [144, 145], addressing resource and environmental constraints through technological innovation.

Key green transformation approaches include:

- (1) Mechanically customizable polyurethane bio-based elastomers to reduce carbon footprint [147].
- (2) Magnetorheological technologies for dynamic stiffness control [148, 149].
- (3) Phase-change material integration in flexible fixtures for thermal management [150, 151].

The progression towards a closed-loop manufacturing ecosystem requires autonomous robotic disassembly technologies. However, technological advancement must align with comprehensive policy frameworks, addressing challenges such as carbon accounting standards and data privacy concerns [146].

5.3 Cross-Domain Integration and Expansion of Emerging Scenarios

The contemporary technological landscape exhibits a complex "technology-policy-application" interaction across space exploration, military technologies, and

robotic mass production [152–154]. This approach enables tailored technological strategies balancing scientific innovation and economic optimization.

Space exploration demands autonomous assembly systems resistant to extreme environments. Utilizing carbon fiber composites and self-healing coating technologies [155], these systems can withstand high radiation, microgravity, and extreme temperatures for lunar base construction.

Battlefield robotics introduced advanced assembly capabilities through:

- (1) Vision-tactile fusion systems with 3D point cloud registration [156]
- (2) Swarm robotics with ant colony optimization [157]
- (3) Federated learning frameworks for energy-efficient multi-robot collaboration [158]

Bionic robot production requires nanometer-level precision in joint actuator assembly [159]. Innovative technologies include:

- (1) Laser-induced breakdown spectroscopy (LIBS) for real-time surface monitoring
- (2) Adaptive impedance control to minimize micro-crack formation
- (3) Multi-modal sensor fusion for electromagnetic compatibility

Additionally, the assembly of bionic robot joints using frameless torque motors can also attempt multi-modal sensor fusion technologies to ensure electromagnetic compatibility and thermal stability. On the production line of bionic humans, augmented reality-assisted assembly systems can guide workers through complex procedures via gesture interaction [160].

6 Conclusions

As a core component of intelligent manufacturing, robot peg-in-hole assembly technology plays a crucial role in determining the competitiveness of strategic industries such as high-end equipment manufacturing, aerospace, and micro-nano electronics. This paper systematically reviews the composition and architecture of peg-in-hole assembly systems, contact dynamics models, compliance control strategies, and emerging technological trends, highlighting the key bottlenecks in current research, including cross-modal perception fusion, dynamic environmental adaptability, and green material applications. PCDs compensate for pose errors through mechanical structure optimization, but they have limitations in terms of versatility and energy efficiency. Active compliance control relies on high-precision environmental modeling,

which restricts its application in unstructured scenarios. Intelligent flexible assembly, while leveraging deep learning and RL to generalize strategies, still faces challenges in bridging the gap between simulation and reality and overcoming data privacy barriers.

Future research should focus on three main directions:

- (1) Developing a virtual-real integration incremental learning framework, combining neural networks and biomimetic mechanism design, to achieve breakthroughs in nanometer-level assembly precision.
- (2) Promoting the innovation of green materials and low-carbon processes, and constructing a full-life-cycle carbon footprint regulation system.
- (3) Deepening cross-domain technology integration, and exploring emerging scenarios such as assembly in extreme space environments, rapid battlefield response systems, and mass production of bionic humans.

Further collaborative innovation is needed to overcome core component autonomy issues, the lack of standardized protocols, and other "bottleneck" challenges. These efforts will drive robot peg-in-hole assembly technology toward intelligent, sustainable, and universal development, injecting new momentum into the global manufacturing industry's transformation and upgrading.

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Author Contributions

Shanglin Li contributed to the conceptualization of the research, performed formal analysis, and drafted the original manuscript. Hao Gong secured project funding, reviewed and edited the manuscript, and developed the methodology. Jianhua Liu conducted investigations and provided supervision. Jiakai Li developed software and created visualizations. Xinjian Deng managed data curation and provided research resources. All authors reviewed and approved the final manuscript.

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Data availability

This review article does not generate new experimental datasets. The paper is based on a comprehensive analysis of existing literature in the field of robotic peg-in-hole assembly. All sources analyzed in this review are properly cited and referenced in the bibliography. The search strategy, inclusion criteria, and complete list of reviewed publications are available in the methodology section. Additional details regarding the systematic review process are available from the corresponding author upon reasonable request.

Declarations

Competing Interests

The authors declare no competing financial interests.

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